A TUPLE BECOMES A DIRECT SUMMAND INCLUSION AFTER INVERTING SOME ELEMENT OF THE COMPLEMENT OF A POINT EXACTLY WHEN ITS IMAGE GENERATES AN INDEPENDENT FAMILY OF THE FIBRE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed. The counted statement is the labeled lemma environment at source lines 38730-38746 (canonical tag 0GV9) and the counted proof is the proof environment at source lines 38748-38786, which immediately follows it and is its own proof.

Editorial note. The statement is formulated inside the situation of the section, which is why it begins by referring to Situation situation-splitting; that label is carried as a navigation stub because the preamble of this card is the shared chapter preamble and the situation environment itself is not counted.

Editorial note. The proof contains one footnote, the Kronecker delta explaining the notation delta_ij, at source line 38761. It is part of the author's own text inside the counted span and is retained.

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Allowed standard prerequisites (editorial).

- A family of elements of a module is the inclusion of a direct summand after localizing at f exactly when there are dual maps whose composite matrix has determinant mapping to a unit of the localization. used at proof lines 38762-38772, where the composite of the tuple with the dual family has determinant f mapping to 1 in kappa(x), which gives the splitting after inverting f. Produces the direct summand condition from the independence of the images in V(x).
- Localization commutes with Hom from a finitely presented module, so maps defined after inverting f can be cleared by multiplying with a power of f. used at proof lines 38778-38783, where the maps phi_i over R_f are replaced by maps phi'_i over R with phi'_i(s_j) = f^n delta_ij. Provides

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the converse direction: the dual family over the point comes from an actual family of maps over R after multiplying by a power of f.

- The perpendicular B(x) of the image of the evaluation map consists of the linear forms vanishing on the images of all R-linear maps, and the quotient V(x) = M(x)/B(x) is the fibre that detects independence. used at proof lines 38749-38756. Fixes the objects in terms of which both directions of the equivalence are formulated.

Source reference. Original lemma 0583: sources/stacks-chow/tags.txt, source lines 6766–6766. The reference supplies the standard compatibility of Hom from a finitely presented module with localization, used at proof lines 38780-38782 to clear the denominators f in the dual maps; the argument that consumes it, namely passing from the maps over R_f to the maps phi'_i over R and reading off independence, is inside the counted span.

Situation 0.2. Here R is a ring and M is a finitely presented R-module. Denote $\Omega \subset \mathrm{Spec}(R)$ the set of closed points with the induced topology. For $x \in \Omega$ denote $M(x) = M/xM$ the fibre of M at x. This is a finite dimensional vector space over the residue field $\kappa(x)$ at x. Given $s \in M$ we denote $s(x)$ the image of s in $M(x)$.

Lemma 0.3. *In Situation 0.2 let $x \in \Omega$. There exists a canonical short exact sequence*

$$0 \to B(x) \to M(x) \to V(x) \to 0$$

of $\kappa(x)$-vector spaces which the following property: for $s_1, \ldots, s_r \in M$ the following are equivalent

1. *there exists an $f \in R$, $f \notin x$ such that the map $s_1, \ldots, s_r : R^{\oplus r} \to M$ becomes the inclusion of a direct summand after inverting f, and*
2. *$s_1(x), \ldots, s_r(x)$ map to linearly independent elements of $V(x)$.*

Proof. Define $B(x) \subset M(x)$ as the perpendicular of the image of the map

$$\mathrm{Hom}_R(M, R) \to \mathrm{Hom}_{\kappa(x)}(M(x), \kappa(x))$$

and set $V(x) = M(x)/B(x)$. Then any R-linear map $\varphi : M \to R$ induces a map $\overline{\varphi} : V(x) \to \kappa(x)$ and conversely any $\kappa(x)$-linear map $\lambda : V(x) \to \kappa(x)$ is equal to $\overline{\varphi}$ for some φ. Let $s_1, \ldots, s_r \in M$.

Suppose $s_1, \ldots, s_r$ map to linearly independent elements of $V(x)$. Then we can find $\varphi_1, \ldots, \varphi_r \in \mathrm{Hom}_R(M, R)$ such that $\varphi_i(s_j)$ maps to δ_{ij}[1] in $\kappa(x)$. Hence the matrix of the composition

$$R^{\oplus r} \xrightarrow{s_1, \ldots, s_r} M \xrightarrow{\varphi_1, \ldots, \varphi_r} R^{\oplus r}$$

has a determinant $f \in R$ which maps to 1 in $\kappa(x)$ Clearly, this implies that $s_1, \ldots, s_r : R^{\oplus r} \to M$ is the inclusion of a direct summand after inverting f.

Conversely, suppose that we have an $f \in R$, $f \notin x$ such that $s_1, \ldots, s_r : R^{\oplus r} \to M$ is the inclusion of a direct summand after inverting f. Hence we can find R_f-linear maps $\varphi_i : M_f \to R_f$ such that $\varphi_i(s_j) = \delta_{ij} \in R_f$. Since $\mathrm{Hom}_R(M, R)_f = \mathrm{Hom}_{R_f}(M_f, R_f)$ by Algebra, Lemma 0583 we conclude that we can find $n \geq 0$ and $\varphi_i' \in \mathrm{Hom}_R(M, R)$ such that $\varphi_i'(s_j) = f^n \delta_{ij} \in R$. It follows that $s_1, \ldots, s_r$ map to linearly independent elements of $V(x)$ as $\overline{\varphi}_i'(s_j) = f^n \delta_{ij}$. □

[1] Kronecker delta.